Lecture 8 Practice Questions and Solutions

Question 1 (q1_sol.py)

Make a Python version of the math function f(x) = 2x + 1. Use it print the values f(2), f(5), and f(1000). Then ask the user to enter a number and print the value of f(x) as shown.

```
5
11
2001
Enter a number: 2.6
f(2.6) = 6.2
```

Sample Solution

```
def f(x):
    return 2 * x + 1

print(f(2))
print(f(5))
print(f(1000))

x = float(input("Enter a number: "))
print(f'f({x}) = {f(x)}')
```

Question 2 (q2_sol.py)

Write a function that takes a person's name as input, and returns a string of the form 'Hello, <name>!'. Then ask the user to enter the names of two people, and print the result of calling the function for each person.

```
Person 1: Audrey
Person 2: Ali

Hello, Audrey!
Hello, Ali!
```

Sample Solution

```
def make_greeting(name):
    return f'Hello, {name}!'

name1 = input("Person 1: ")
name2 = input("Person 2: ")
print()
print(make_greeting(name1))
print(make_greeting(name2))
```

Question 3 (q3_sol.py)

Write a function called print_in_bars(word) that prints word surrounded by | characters as shown. This function only prints, and so has no return. Then ask the user to enter their three favourite foods, and use print_in_bars to print each food.

```
Food 1: peaches
Food 2: sushi
Food 3: ice cream

|peaches|
|sushi|
|ice cream|
```

Sample Solution

```
def print_in_bars(word):
    print(f'|{word}|')

food1 = input("Food 1: ")
food2 = input("Food 2: ")
food3 = input("Food 3: ")
print()
print_in_bars(food1)
print_in_bars(food2)
print_in_bars(food3)
```

Question 4 (q4_sol.py)

Write a function called print_in_box(word) that prints word in a box as shown. Use the print_in_bars function from the previous question to print the middle line of the box. Then ask the user to enter a movie name and print in a box.

```
Enter a movie name: The Matrix
+----------+
|The Matrix|
+----------+
```

Sample Solution

```
def print_in_bars(word):
    print(f'|{word}|')

def print_in_box(word):
    bar = '+' + '-' * len(word) + '+'
    print(bar)
    print_in_bars(word)
    print(bar)


movie = input("Enter a movie name: ")
print_in_box(movie)
```

Question 5 (q5_sol.py)

Write a program that asks the user to enter an int that is 0 or greater, and a word, and then prints the word in a numbered list as shown. Use a for-loop in your solution. No functions are required, but you can make one if you want.

```
Enter an int: 3
Enter a word: hello!
hello!
hello!
hello!
```

Sample Solution

```
n = int(input("Enter an int: "))
word = input("Enter a word: ")
print()
for i in range(n):
    print(f'{i+1}. {word}')
```

Question 6 (q6_sol.py)

Write a program that asks the user to enter an int (called n)that is 0 or greater. Then, starting at 1, print the ints from 1 to n on the screen in a box as shown. Use the print_in_box(word) function from the previous question.

```
Enter an int: 3

+-+
|1|
+-+

+-+
|2|
+-+

+-+
|3|
+-+
```

Sample Solution

```
def print_in_bars(word):
    print(f'|{word}|')

def print_in_box(word):
    bar = '+' + '-' * len(word) + '+'
    print(bar)
    print_in_bars(word)
    print(bar)

n = int(input("Enter an int: "))
print()
for i in range(1, n+1):
    print_in_box(str(i))
    print()
```